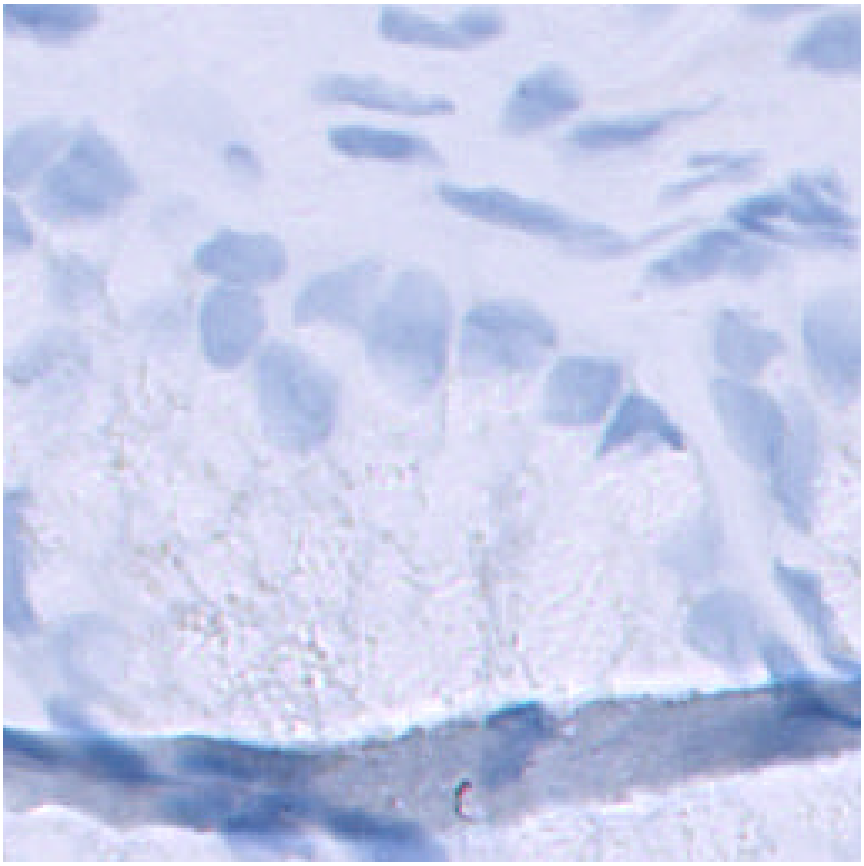
Deep Learning for the Detection of *Helicobacter pylori* in Immunohistochemical Histopathological Images

Project Report



Representative histology: IHC-stained gastric biopsy patch.

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Abstract

We use the Quiron Salut dataset of 245 gastric biopsies (immunohistochemically stained whole-slide images) to implement and compare two patch-level models for *Helicobacter pylori* detection. A patch-based pipeline processes 256×256 patches extracted along tissue borders. Two approaches are implemented: (i) a supervised convolutional neural network (CNN) trained on expert-annotated patches, and (ii) an unsupervised autoencoder (AE) trained on healthy patches and used for anomaly detection via reconstruction error. Both models are evaluated at patch level with 5-fold patient-level cross-validation and at patient level by aggregating patch predictions into a contamination ratio and applying a learned threshold; we report results on the cross-validation splits and on the independent Hold-out set (116 patients). The CNN achieves stronger and more stable performance (patch-level F1 0.92, patient-level CV F1 0.95, Hold-out F1 0.81); the AE is competitive at patch level but shows higher variability and lower specificity at patient level, with one fold exhibiting a pronounced reconstruction-error scale shift.

Contents

Glossary 2

1 Introduction 3

2 Dataset Description 3

3 Methodology 3
- 3.1 Pipeline overview . . . 3
- 3.2 Preprocessing . . . 5
- 3.3 Patch-level approach 1: CNN classifier . . . 5
- 3.4 Patch-level approach 2: Autoencoder anomaly detection . . . 5
- 3.5 Patient-level aggregation . . . 5

4 Implementation Details 6
- 4.1 Development iterations . . . 6
- 4.2 Code structure . . . 6

5 Experimental Design 7
- 5.1 Hyperparameters . . . 7

6 Results 7
- 6.1 Patch-level metrics . . . 7
- 6.2 Cross-validation variability and AE fold anomaly . . . 8
- 6.3 Patient-level . . . 10
- 6.4 Patient-level visualisations . . . 10

7 Discussion 14

8 Conclusion 14

References 15

A Additional AE fold diagnostics 15

B Code extracts 15

Glossary

AE Autoencoder. Unsupervised model trained on healthy patches; anomaly score is reconstruction error.

CNN Convolutional neural network. Supervised patch-level classifier (healthy vs. bacteria).

Contamination ratio For a patient, the fraction of patches predicted positive; $r_p = \frac{1}{N_p} \sum_{i=1}^{N_p} \hat{z}_i$.

CV Cross-validation. Here: 5-fold patient-level CV (splits by patient, not by patch).

F1 F1 score. Harmonic mean of precision and recall: $2 \cdot \text{precision} \cdot \text{recall}/(\text{precision} + \text{recall})$.

FPR False positive rate. Proportion of true negatives incorrectly predicted positive; $\text{FP}/(\text{TN} + \text{FP})$.

Hold-out Independent test set (116 patients), not used for training or threshold selection; evaluated only with trained models.

IHC Immunohistochemistry. Staining used on the gastric biopsy slides.

Patch A 256×256 pixel crop from a whole-slide image (here, along the tissue border).

Patch-level Evaluation or model applied to individual patches (as opposed to a full patient).

Patient-level Evaluation or prediction after aggregating patch outputs per patient (e.g. via contamination ratio and $\tau_{patient}$).

Precision Among predicted positives, the fraction that are truly positive; $\text{TP}/(\text{TP} + \text{FP})$.

Recall Among true positives, the fraction correctly predicted positive; $\text{TP}/(\text{TP} + \text{FN})$. Syn. sensitivity.

ROC Receiver operating characteristic. Curve of true positive rate vs. false positive rate as the decision threshold varies.

Specificity Among true negatives, the fraction correctly predicted negative; $\text{TN}/(\text{TN} + \text{FP})$.

TPR True positive rate. Same as recall; $\text{TP}/(\text{TP} + \text{FN})$.

$\tau_{patient}$ Patient-level threshold. A patient is predicted positive if their contamination ratio $r_p > \tau_{patient}$; chosen per fold on training patients (best F1).

τ_{patch} Patch-level threshold. A patch is predicted positive if its score (CNN probability or AE reconstruction error) exceeds τ_{patch}; for the AE, chosen per fold on the validation set (ROC).

WSI Whole-slide image. High-resolution digitised histology slide.

1 Introduction

Helicobacter pylori is a gastric bacterium associated with chronic gastritis and an increased risk of gastric cancer. In clinical practice, detection often relies on the visual inspection of immunohistochemically (IHC) stained histopathology slides. Whole-slide images (WSI) can exceed 100 000 pixels per side, making manual review time-consuming. This project investigates automated, patch-based detection: we classify small image patches as healthy or containing bacteria, then aggregate patch predictions to obtain a patient-level diagnosis.

The pipeline has three stages: (1) train a patch-level classifier on expert-annotated data; (2) apply it to all patches of each patient and compute the fraction of positive patches (contamination ratio); (3) compare that ratio to a threshold $\tau_{patient}$ to yield a binary patient-level diagnosis. Key challenges are the limited number of annotated patches, strong class imbalance (few patches contain bacteria), and the need to avoid data leakage by splitting at the patient level rather than the patch level.

We implement two patch-level models—a supervised CNN and an unsupervised autoencoder used for anomaly detection—and evaluate the full pipeline: 5-fold patient-level cross-validation for patch-level training, then patient-level aggregation with fold-consistent models and thresholds, evaluated on both the CV validation folds and the independent Hold-out set (116 patients). This report presents the methodology, implementation, experimental design, and results for the complete pipeline.

2 Dataset Description

We use the Quiron Salut dataset: IHC-stained gastric biopsy whole-slide images from which 256×256 pixel patches have been cropped along the tissue border. Patient-level labels indicate H. *pylori* density: NEGATIVE, LOW, or HIGH. For binary classification we map NEGATIVE to 0 and (LOW, HIGH) to 1.

Patch-level annotations are provided in `CoordAnnotatedAllPatches.csv` with a `Presence` field: -1 (healthy), 1 (bacteria present), or 0 (unclear, which we excluded from training as there are 3 or 4 cases only).

The data are organised into three parts: **CrossValidation/Annotated** (expert-labelled patches for training and validating patch-level models), **CrossValidation/Cropped** (all patches per patient for aggregation experiments), and **Hold-out** (116 independent patients reserved for final evaluation only). Table 1 summarises the sample counts.

Split	#Patients	#Annotated patches	#Positive patches	#Negative patches
CrossValidation	123	1211	161	1050
Hold-out	116	–	–	–

Table 1: Dataset size overview (Quiron Salut).

3 Methodology

3.1 Pipeline overview

The pipeline has three steps. First, a patch-level model (CNN or autoencoder) is trained on annotated or healthy-only patches, with 5-fold patient-level cross-validation so that each fold yields a trained model and (for the AE) a patch-level threshold. Second, each fold model is applied to every patch of each patient (CV and Hold-out); the fraction of patches predicted positive per patient defines the contamination ratio r_p. Third, a patient-level threshold $\tau_{patient}$ is chosen per fold on the fold's training patients (maximising F1), then applied to validation

and Hold-out patients to obtain binary patient-level predictions. We compare the CNN and the autoencoder at both patch level and patient level, reporting metrics on the CV folds and on the independent Hold-out set. Figure 1 summarises the workflow.

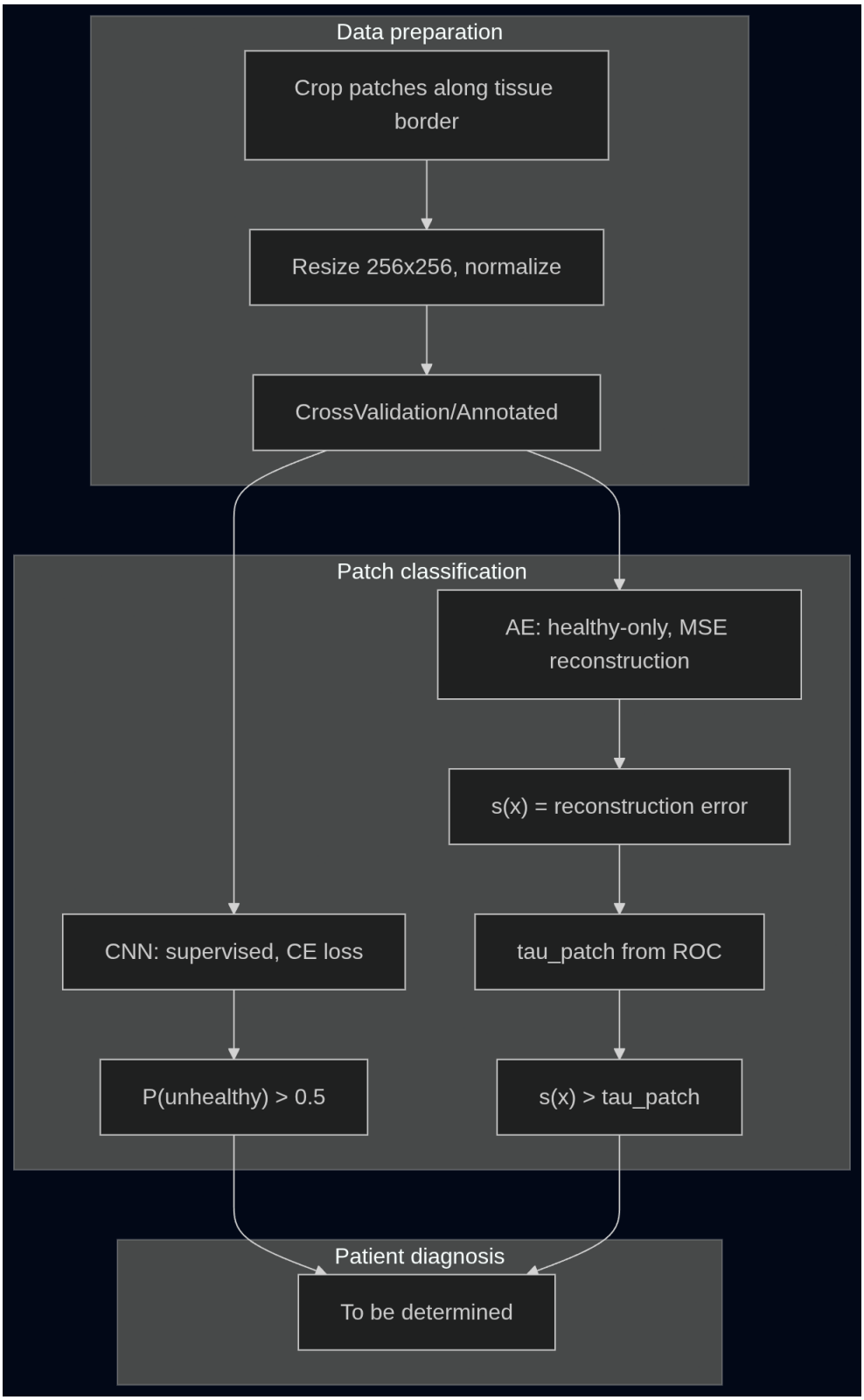


Figure 1: Methodology workflow: data preparation, patch-level classification (CNN vs autoencoder), and patient-level aggregation evaluated on CV and Hold-out.

3.2 Preprocessing

Patches are loaded as RGB images, resized to 256×256 if needed, and normalised per channel with $x' = (x - \mu)/\sigma$ where $\mu = \sigma = (0.5, 0.5, 0.5)$, so that inputs lie approximately in $[-1, 1]$. For the CNN we apply data augmentation: random flips, small rotations, and optional colour jitter.

3.3 Patch-level approach 1: CNN classifier

We use a lightweight CNN with three convolutional blocks followed by fully connected layers. Training minimises cross-entropy loss. With one-hot label y and predicted class probabilities $\hat{p}_c$ (softmax over logits), the loss is

$$\mathcal{L}_{\text{CE}} = -\sum_{c \in \{0,1\}} y_c \log \hat{p}_c,$$

with $c = 0$ for healthy and $c = 1$ for patches containing H. *pylori*. The model is trained with Adam; early stopping is optional. At inference, a patch is predicted positive if the probability of class 1 exceeds 0.5.

3.4 Patch-level approach 2: Autoencoder anomaly detection

The autoencoder is trained only on healthy patches and learns to reconstruct normal tissue. Training minimises mean squared error between input and reconstruction:

$$\mathcal{L}_{\text{AE}} = \frac{1}{CHW} \sum_{c,i,j} (x_{c,i,j} - \hat{x}_{c,i,j})^2,$$

where $C = 3$, $H = W = 256$. The per-patch anomaly score $s(x)$ is the reconstruction error. We use either the mean over all pixels or the maximum over 8×8 spatial cell means (`max_local`) so that small anomalous regions contribute more. A patch is predicted positive if $s(x) > \tau_{patch}$. The threshold τ_{patch} is chosen on the validation set as the point on the ROC curve closest to (FPR= 0, TPR= 1), implemented as `roc_threshold_optimal`.

3.5 Patient-level aggregation

For each patient p with N_p patches and binary patch predictions $\hat{z}_i \in \{0, 1\}$, we define the contamination ratio

$$r_p = \frac{1}{N_p} \sum_{i=1}^{N_p} \hat{z}_i.$$

The patient is predicted positive if $r_p > \tau_{patient}$. In our implementation, $\hat{z}_i$ is derived from a patch-level score and threshold:

- CNN: score = $P(\text{unhealthy})$, threshold $\tau_{patch}^{cnn} = 0.5$ (default).
- AE: score = reconstruction error (`mean` or `max_local`), threshold τ_{patch}^{ae} from ROC (`autoencoder_threshold.txt` or fold-specific threshold files).

The patient threshold $\tau_{patient}$ is selected on training patients by maximising patient-level F1 over candidate ratio thresholds, then evaluated on validation patients. The same protocol is applied fold-by-fold in patient-level CV, and then on Hold-out.

4 Implementation Details

4.1 Development iterations

The CNN started as a minimal classifier: three small convolutional blocks (3→16→32→64 channels, ReLU, max pooling) and two fully connected layers mapping the flattened 64×32×32 features to two class logits. No architectural changes were required for good performance; the main refinements were patient-level train/validation splitting to avoid leakage, ignoring patches with `Presence=0` in the annotation file, and training for more epochs (e.g. 150 in the Slurm script). The current design remains lightweight and fast to train.

The autoencoder began with a shallower encoder (32→64→128→256→256 channels, 8×8 bottleneck) and ReLU activations. To avoid losing information in the bottleneck (ReLU zeros negatives), ReLU was replaced by LeakyReLU. The encoder and decoder were then deepened and widened (64→128→256→512→512 channels, 8×8×512 bottleneck) to improve reconstruction of healthy tissue so that anomalous patches yield higher error. The anomaly score was extended with a `max_local` option (maximum over 8×8 spatial cell means) so that focal H. *pylori* regions drive the score. Checkpoint selection was fixed to keep the best model by validation loss instead of the last epoch, and the default aggregation and epoch count were set to `max_local` and 35 (overridable via Slurm).

4.2 Code structure

Core scripts:

- `model_classifier.py`: CNN definition, training, and evaluation.
- `model_autoencoder.py`: AE training and ROC-based patch threshold selection.
- `src/helico/cv_utils.py`: shared patient-level split and metric helpers.
- `compare_patch_classifiers.py`: side-by-side patch-level comparison.

Patient-level diagnosis is implemented in `patient_level_pipeline.py`. The script supports:

- single-checkpoint mode;
- fold-consistent mode (`--use-cv-fold-checkpoints`) with models from `cv/cnn/fold_i/` and `cv/autoencoder/fold_i/`.

Outputs are written under `output/patient_level/`: fold metrics, Hold-out-per-fold metrics, patient-level detail rows, and summary statistics.

Cluster scripts:

- training: `train_classifier.sbatch.sh`, `train_autoencoder.sbatch.sh` (legacy `train.sbatch.sh`);
- patient-level: `patient_cnn_fold.sbatch.sh`, `patient_autoencoder_fold.sbatch.sh`, `patient_plots.sbatch.sh`;
- diagnostics:
  - `analyze_cv_metrics.py`
  - `analyze_ae_fold_diagnostics.py`
  - `analyze_patient_level_metrics.py`

5 Experimental Design

We split data at the **patient level** so that a patient never appears in both train and validation folds. The current evaluation uses **5-fold patient-level cross-validation** (`--k-folds 5`, `seed=42`) for both CNN and AE. For each fold, checkpoints and metrics are stored under `cv/cnn/fold_i/` and `cv/autoencoder/fold_i/`. The Hold-out set (116 patients) remains reserved for a single final evaluation after all model and threshold design choices are fixed.

For patient-level diagnosis, we compute contamination ratios on `CrossValidation/Cropped` and select fold-wise $\tau_{patient}$ from training patients only (best F1). We then evaluate on validation patients and additionally report Hold-out metrics per fold using the same fold model and $\tau_{patient}$. This avoids mixing fold artefacts and keeps patch-level and patient-level protocols consistent.

We report fold-wise and aggregate patch-level metrics (accuracy, precision, recall, specificity, F1), using both mean±std and robust summaries (median/IQR). For AE we additionally report threshold behaviour, TPR and FPR by fold.

5.1 Hyperparameters

Hyperparameter	Value
Optimizer	Adam
Learning rate	10^{-3}
Batch size	32
Epochs (CNN)	150
Epochs (AE)	150
Weight decay	0 (not used)
Early stopping	not used
CV protocol	patient-level 5-fold, `seed=42`
Patch threshold τ_{patch}	ROC closest-to-(0,1) criterion (`roc_threshold_optimal`)
Patient threshold $\tau_{patient}$	per fold from CV train (best F1); applied to CV val and Hold-out

Table 2: Hyperparameters and training configuration (from `model_classifier.py`, `model_autoencoder.py`, and Slurm scripts).

6 Results

Both models were evaluated with patient-level 5-fold CV. Aggregate results are reported in Table 3. Figure 2 shows mean±std comparison across folds. For AE, fold-level diagnostics revealed one anomalous fold (fold 3) with an error-scale shift and much higher optimal threshold; this is detailed in Figure 4.

6.1 Patch-level metrics

Table 3 summarises patch-level performance. Recall is TP/(TP+FN) and specificity is TN/(TN+FP). The CNN attains higher accuracy and precision; the autoencoder is slightly lower but still strong, and was trained without any positive patch labels.

Method	Accuracy	Precision	Recall	Specificity	F1
CNN ($\mu \pm \sigma$)	0.9108±0.0484	0.9706±0.0197	0.8783±0.0717	0.9532±0.0499	0.9201±0.0382
Autoencoder ($\mu \pm \sigma$)	0.8246±0.0736	0.8836±0.0619	0.8065±0.1009	0.8563±0.0380	0.8397±0.0696

Table 3: Patch-level cross-validation performance (5-fold patient-level CV).

6.2 Cross-validation variability and AE fold anomaly

Figure 2 compares fold-aggregated performance between models. CNN remains stronger and more stable across folds. AE is competitive but more sensitive to fold composition and reconstruction-error scale shifts.

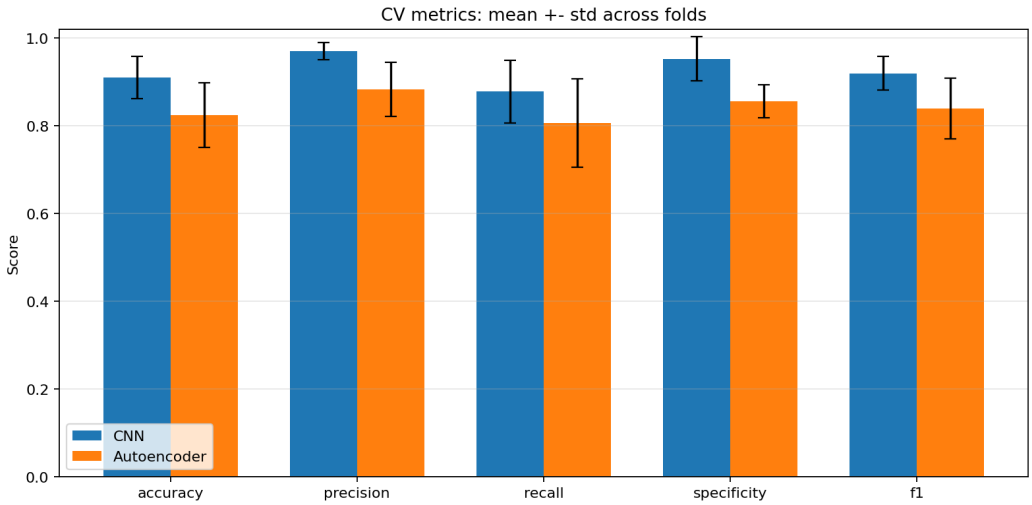


Figure 2: Model comparison across patient-level CV folds (mean±std).

Figure 3 shows the same patch-level metrics as box plots (median and IQR across the 5 folds), giving a robust view less influenced by outliers than mean±std.

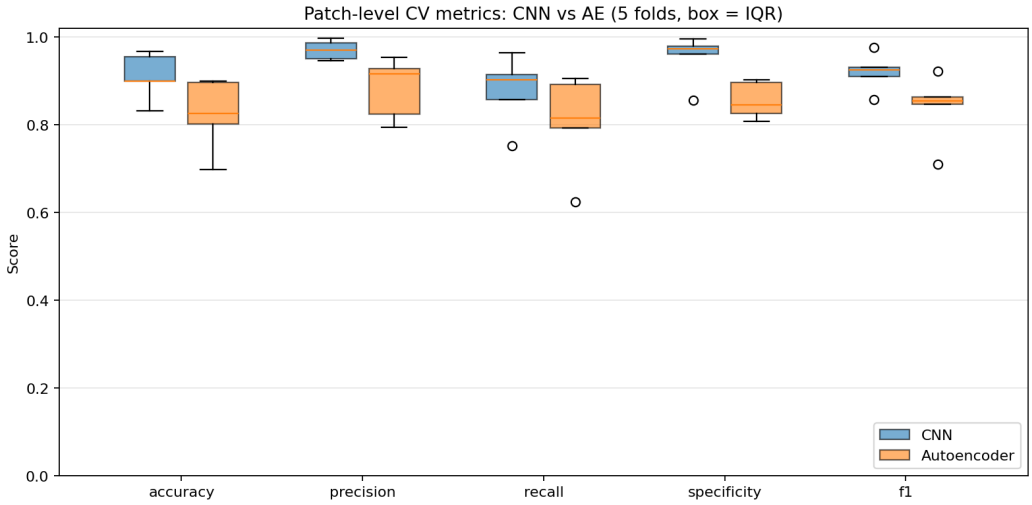


Figure 3: Patch-level metrics by fold: CNN vs AE (box = IQR, 5 folds).

AE fold diagnostics (Figure 4) show fold 3 has a distinct error scale compared with folds 1, 2, 4, and 5. The fold-3 local threshold is 0.7871 while other folds are around 0.04–0.07. Class-wise reconstruction-error medians in fold 3 are also much larger (healthy median ≈0.60, unhealthy median ≈1.05), indicating a distribution shift rather than a simple class-imbalance artifact. A full five-fold visual comparison is provided in Appendix Figure 14.

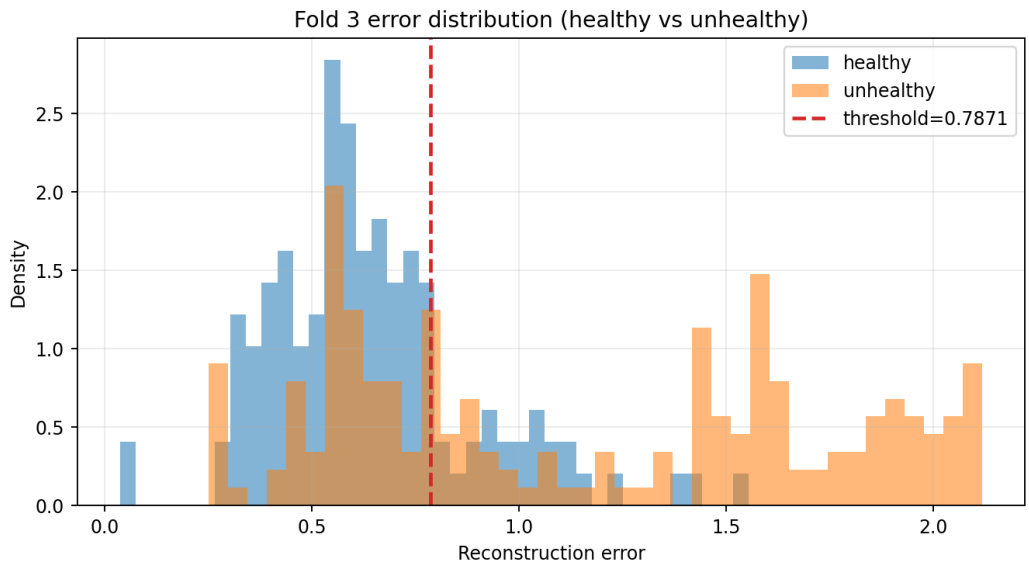
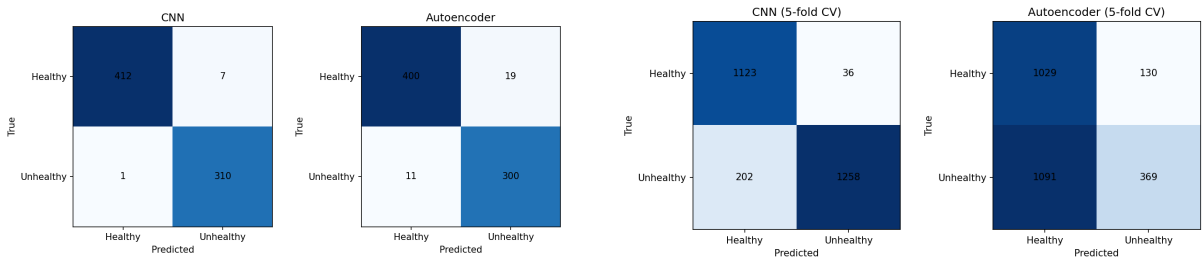


Figure 4: Fold-3 AE reconstruction-error distributions (healthy vs unhealthy) and selected threshold.

Robust summaries confirm that mean/std is distorted by outliers. For AE threshold, mean±std is 0.2092±0.2892 but the robust estimate is median 0.0717 with IQR 0.0027, showing that one outlier fold dominates the standard deviation. Table 4 gives the patch-level threshold τ_{patch}, TPR and FPR per fold; fold 3 is the clear outlier.

Fold	τ_{patch} (AE)	TPR	FPR
1	0.0436	0.818	0.154
2	0.0717	0.911	0.103
3	0.7871	0.635	0.192
4	0.0705	0.796	0.173
5	0.0732	0.897	0.096

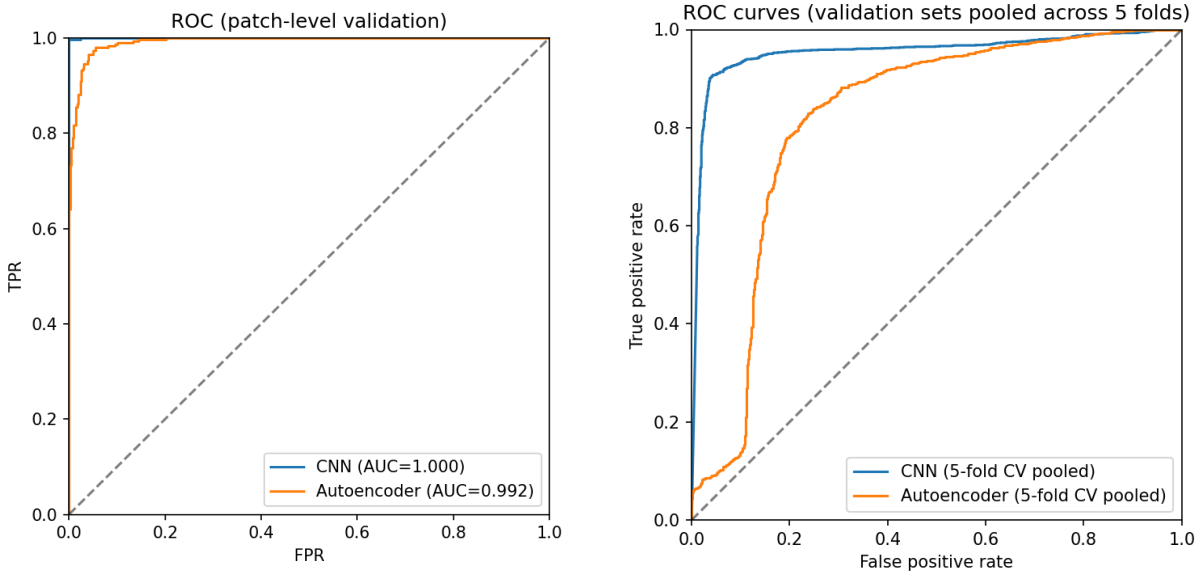
Table 4: Autoencoder patch-level threshold and operating point per CV fold (from `cv/autoencoder/fold_*/metrics.json`). Fold 3 exhibits a much higher τ_{patch} due to error-scale shift.



(a) Single run (one train/val split).

(b) 5-fold CV (validation sets pooled).

Figure 5: Patch-level confusion matrices (single run: one train/val split; when available, right panel shows 5-fold CV pooled). Left panel: CNN. Right: Autoencoder (threshold from validation ROC or mean of fold thresholds).



(a) Single run (one train/val split).

(b) 5-fold CV (scores pooled across folds).

Figure 6: ROC curves (single run; when available, right panel shows 5-fold CV pooled scores).

6.3 Patient-level

Patient-level performance is computed by `patient_level_pipeline.py` in fold-consistent mode (`--use-cv-fold-checkpoints`). For each fold, the corresponding patch model is loaded, contamination ratios are computed, and $\tau_{patient}$ is fitted on fold-train patients (best F1), then applied to fold-validation and Hold-out patients. The pipeline stores fold CV metrics and Hold-out-per-fold metrics in JSON files under `output/patient_level/`. Plots and text summaries are generated by `analyze_patient_level_metrics.py`.

Split / Model	Accuracy	Precision	Recall	Specificity	F1
CV / CNN	0.9423±0.0170	0.9241±0.0519	0.9749±0.0308	0.9075±0.0510	0.9472±0.0190
CV / AE	0.7744±0.0556	0.7518±0.0308	0.8960±0.0446	0.5995±0.1748	0.8165±0.0224
Hold-out / CNN	0.8172±0.0065	0.8439±0.0173	0.7793±0.0129	0.8552±0.0207	0.8101±0.0053
Hold-out / AE	0.6966±0.0675	0.6860±0.0750	0.7586±0.0218	0.6345±0.1516	0.7172±0.0383

Table 5: Patient-level performance from fold-consistent aggregation (`patient_summary.json` for CNN and AE).

CNN is stronger and more stable at patient level, while AE keeps high sensitivity but lower specificity. This is consistent with patch-level observations and AE fold-threshold variability.

6.4 Patient-level visualisations

The following figures are produced by `analyze_patient_level_metrics.py` (run after the patient-level pipeline with `--use-cv-fold-checkpoints`). They compare CNN and AE at patient level and show threshold behaviour and score distributions.

First, a direct comparison: the bar chart below summarises how the two models perform on the cross-validation folds when we aggregate patches to a patient-level decision. CNN clearly leads on accuracy, precision and F1.

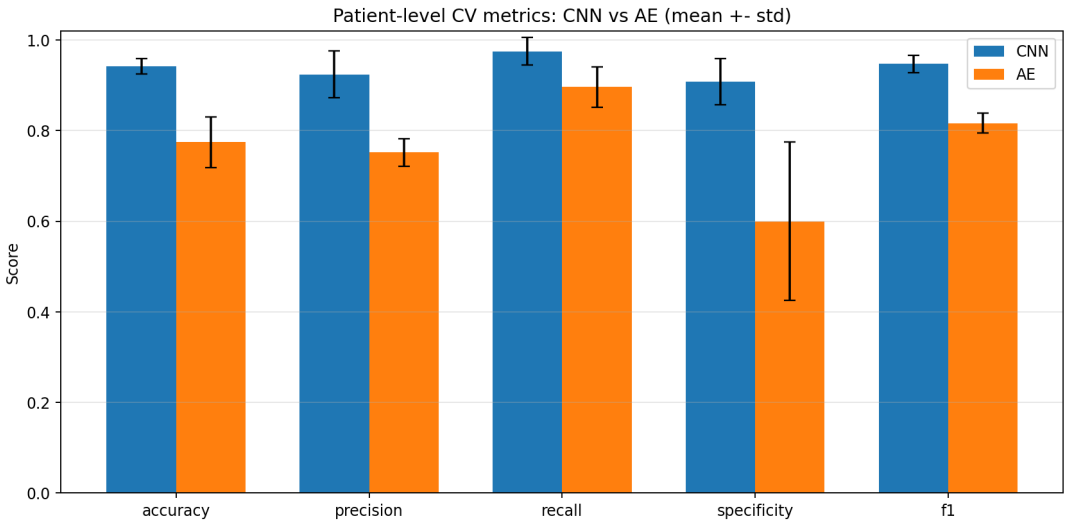
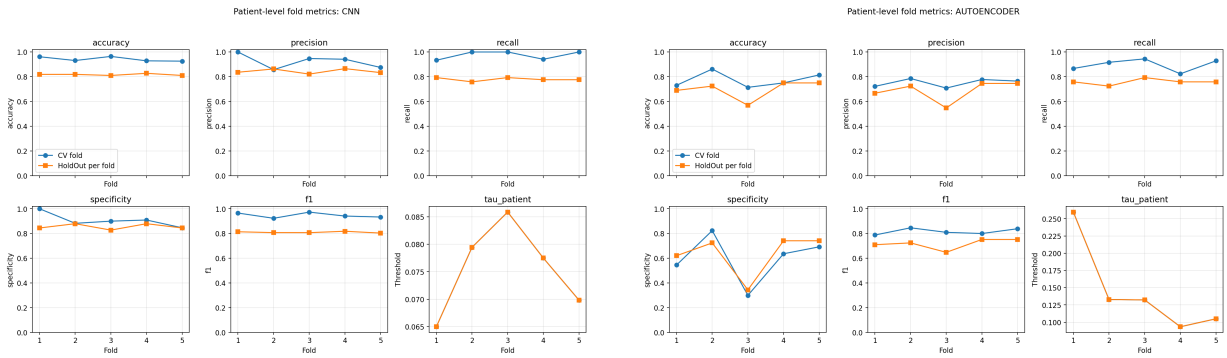


Figure 7: Patient-level CV metrics: CNN vs AE (mean±std across folds).

To see how stable each model is across folds and how performance changes from CV to the unseen Hold-out set, we plot each metric by fold. Circles are the CV validation performance; squares are Hold-out performance using the same fold model. If the two markers track each other, the model generalises well.

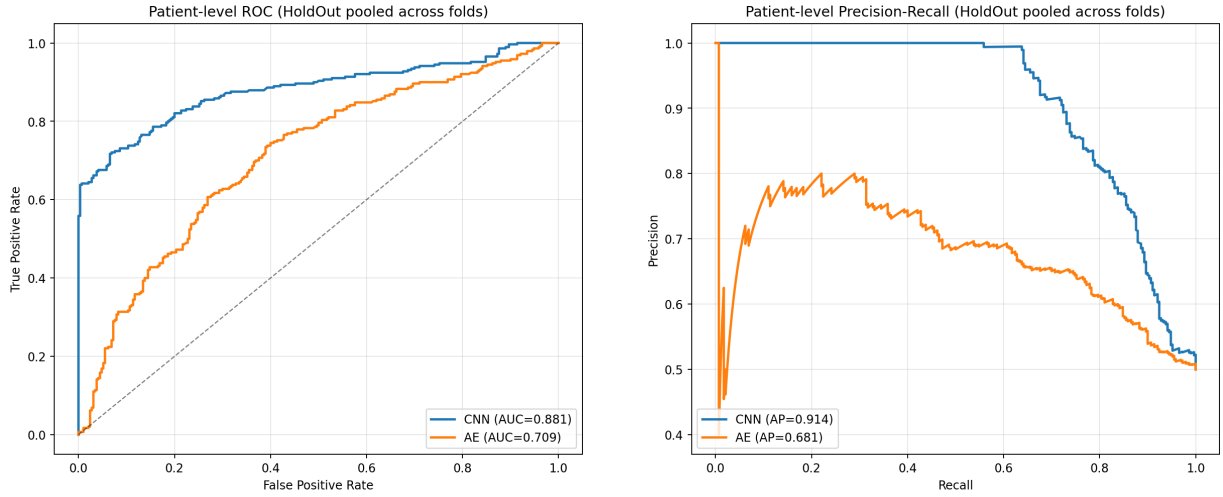


(a) CNN

(b) Autoencoder

Figure 8: Patient-level metrics per fold: CV (circles) vs Hold-out evaluated with the same fold model (squares).

On the independent Hold-out set we can compare the two models in terms of discriminative ability. The ROC and Precision-Recall (PR) curves below show that the CNN again dominates: it achieves both a higher true-positive rate for a given false-positive rate, and higher precision across recall levels.

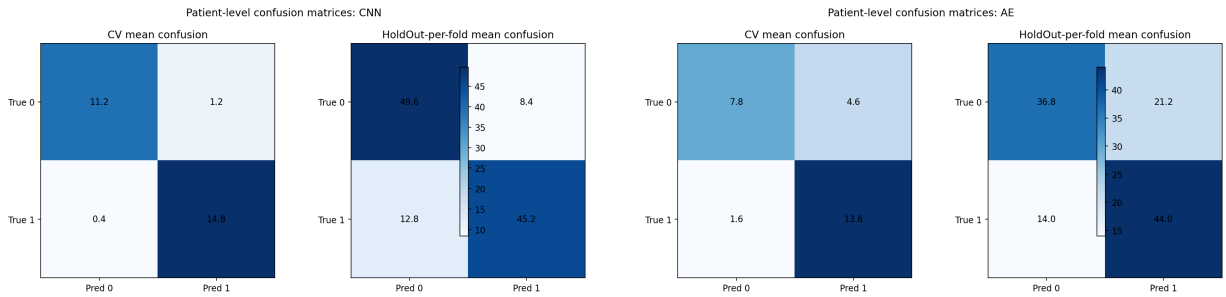


(a) ROC Curve

(b) PR Curve

Figure 9: Patient-level ROC and Precision-Recall curves on Hold-out (pooled across folds).

The next two panels show where the models go wrong. The confusion matrices (averaged over folds) tell us how many patients are correctly or incorrectly classified as healthy or positive, both on CV and on Hold-out. The CNN keeps a better balance; the AE tends to call more patients positive, which boosts sensitivity but hurts specificity.

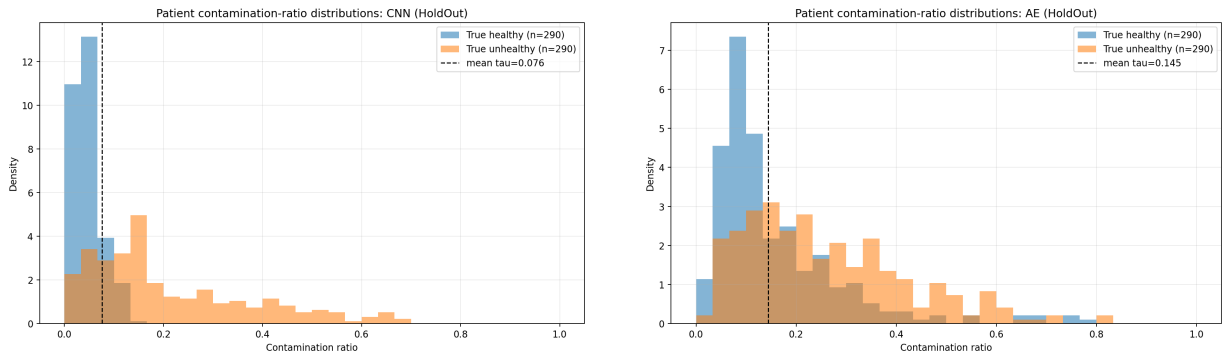


(a) CNN

(b) Autoencoder

Figure 10: Patient-level mean confusion: CV vs Hold-out per fold.

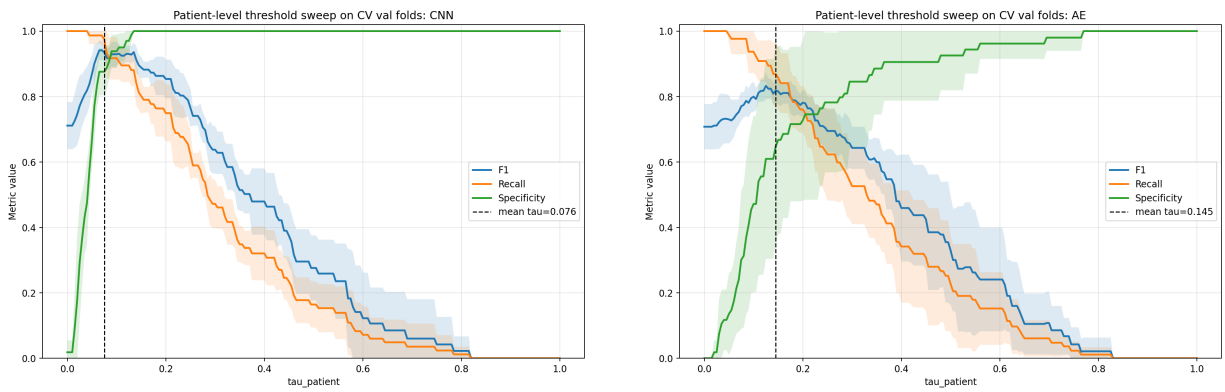
For each patient the model produces a contamination ratio (the fraction of patches predicted positive). Ideally, true healthy patients have low ratios and true positive patients have high ratios, with a clear gap. The histograms below show how these ratios are distributed on Hold-out; the vertical line is the threshold we use for the final decision. Good separation between the two curves means the threshold has an easy job.



(a) CNN (b) Autoencoder

Figure 11: Contamination-ratio distributions on Hold-out (true healthy vs true positive), with mean $\tau_{patient}$ (vertical line).

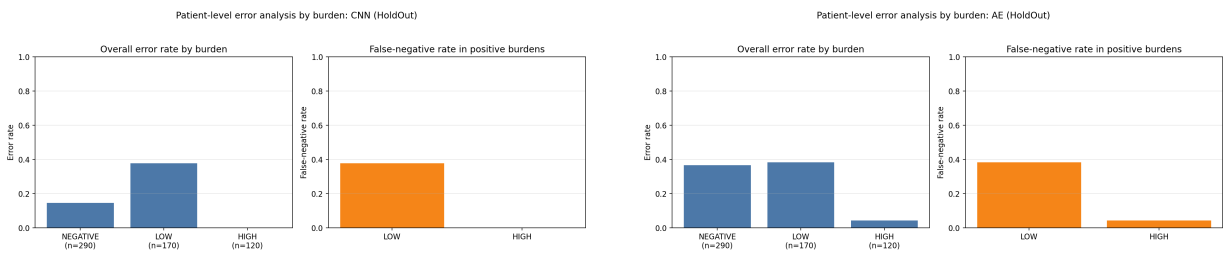
The patient-level threshold $\tau_{patient}$ was chosen on the training set to maximise F1. The curves below show how F1, recall and specificity would change if we moved that threshold: we see the usual trade-off between catching more positives (recall) and avoiding false alarms (specificity).



(a) CNN (b) Autoencoder

Figure 12: F1, recall and specificity vs $\tau_{patient}$ on CV validation folds (mean±std band).

Lastly, we break down errors by the true clinical burden (NEGATIVE, LOW, HIGH). This tells us whether the model fails mainly on low-burden cases, which are harder to detect, or whether mistakes are spread across groups.



(a) CNN (b) Autoencoder

Figure 13: Patient-level error analysis on Hold-out: overall error rate and false-negative rate by burden (NEGATIVE, LOW, HIGH).

7 Discussion

With patient-level 5-fold CV, the CNN consistently outperforms the AE in accuracy/precision/F1 and exhibits lower fold-to-fold variance. The AE remains useful in low-label settings but is notably more sensitive to fold-specific data distribution.

The main observed limitation is threshold instability under distribution shift: fold 3 presents an upward shift of reconstruction errors for both classes, which inflates local threshold and degrades specificity at patient level. A pooled global threshold improved some folds but harmed specificity severely on fold 3, so a single global threshold is not yet robust enough. The current fold-consistent patient pipeline makes this behaviour explicit and reproducible, and provides a solid baseline for future threshold normalisation/calibration strategies (e.g., robust scaling or nested calibration).

8 Conclusion

We implemented and evaluated a full pipeline for H. *pylori* detection in IHC-stained gastric biopsies: two patch-level models (supervised CNN and unsupervised autoencoder for anomaly detection), trained with 5-fold patient-level cross-validation, and a patient-level aggregation stage that computes contamination ratios and applies fold-specific thresholds, evaluated on both the CV validation folds and the independent Hold-out set.

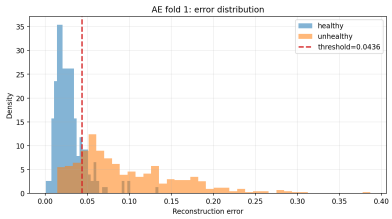
At patch level, the CNN achieves stronger and more stable performance (F1 0.9201 ± 0.0382) than the AE (F1 0.8397 ± 0.0696). At patient level, the CNN again outperforms (CV F1 0.947, Hold-out F1 0.81), while the AE maintains high sensitivity but lower specificity (CV F1 0.82, Hold-out F1 0.72). AE fold diagnostics identify a fold-specific error-scale anomaly (fold 3), motivating robust threshold calibration rather than a single pooled threshold.

Possible continuation. Several directions could extend this work: (i) **Threshold calibration for the AE**—e.g. robust scaling or per-fold normalisation of reconstruction errors so that a single or more stable threshold can be used across folds and specificity improves without losing sensitivity; (ii) **More annotated data**—the current patch set is limited and imbalanced; additional expert labels would help both the CNN and the AE; (iii) **Architecture and training**—deeper or pre-trained patch encoders (e.g. ResNet, vision transformers), and optional class weighting or focal loss to handle imbalance;

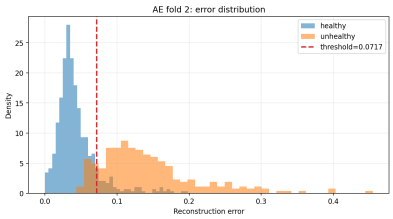
References

1. Anomaly detection via reconstruction: https://arxiv.org/abs/2309.16053 (self-supervised / generative models for small sample size).

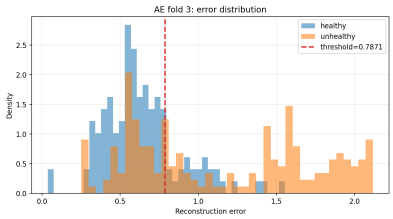
A Additional AE fold diagnostics



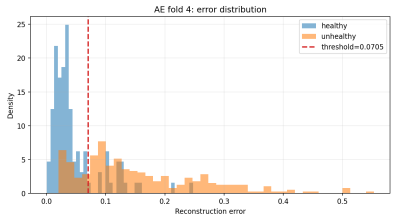
(a) Fold 1



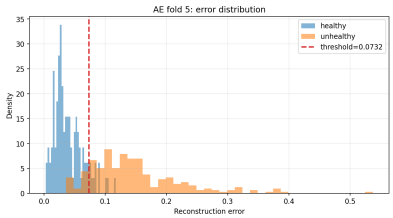
(b) Fold 2



(c) Fold 3



(d) Fold 4



(e) Fold 5

Figure 14: AE reconstruction-error distributions (healthy vs unhealthy) and selected threshold across all 5 folds. Fold 3 exhibits the strongest right-shifted error scale and threshold.

B Code extracts

Listing 1: Comparison script: both models on the same validation set (`compare_patch_classifiers.py`).

```
    # CNN
    cnn = SimpleCNN(num_classes=2).to(device)
    cnn.load_state_dict(torch.load(args.cnn_checkpoint, ...))
    cnn_probs, labels = compute_cnn_probs(cnn, val_loader, device)
    cnn_pred = (cnn_probs >= 0.5).astype(np.int64)

    # Autoencoder
    ae = PatchAutoencoder().to(device)
    ae.load_state_dict(torch.load(args.ae_checkpoint, ...))
    ae_errors, ae_labels = compute_reconstruction_errors(ae, val_loader, device, ...)
    ae_pred = (ae_errors > ae_threshold).astype(np.int64)

    acc_cnn, prec_cnn, cm_cnn = _metrics(labels, cnn_pred)
    acc_ae, prec_ae, cm_ae = _metrics(labels, ae_pred)
```

Listing 2: CNN for patch classification (`model_classifier.py`).

```
class SimpleCNN(nn.Module):
    def __init__(self, num_classes: int = 2) -> None:
        super().__init__()
        self.features = nn.Sequential(
            nn.Conv2d(3, 16, 3, padding=1), nn.ReLU(), nn.MaxPool2d(2),
            nn.Conv2d(16, 32, 3, padding=1), nn.ReLU(), nn.MaxPool2d(2),
```

```
            nn.Conv2d(32, 64, 3, padding=1), nn.ReLU(), nn.MaxPool2d(2),
        )
        self.classifier = nn.Sequential(
            nn.Flatten(), nn.Linear(64 * 32 * 32, 128), nn.ReLU(),
            nn.Linear(128, num_classes),
        )
    def forward(self, x): return self.classifier(self.features(x))
```

Listing 3: Autoencoder: Encoder, Decoder, and full model (`model_autoencoder.py`).

```
class Encoder(nn.Module):
    def __init__(self):
        super().__init__()
        self.conv = nn.Sequential(
            nn.Conv2d(3, 64, 3, padding=1), nn.LeakyReLU(0.2), nn.MaxPool2d(2),
            nn.Conv2d(64, 128, 3, padding=1), nn.LeakyReLU(0.2), nn.MaxPool2d(2),
            nn.Conv2d(128, 256, 3, padding=1), nn.LeakyReLU(0.2), nn.MaxPool2d(2),
            nn.Conv2d(256, 512, 3, padding=1), nn.LeakyReLU(0.2), nn.MaxPool2d(2),
            nn.Conv2d(512, 512, 3, padding=1), nn.LeakyReLU(0.2), nn.MaxPool2d(2),
        )
    def forward(self, x): return self.conv(x)

class Decoder(nn.Module):
    def __init__(self):
        super().__init__()
        self.up = nn.Sequential(
            nn.ConvTranspose2d(512, 512, 2, stride=2), nn.LeakyReLU(0.2),
            nn.ConvTranspose2d(512, 256, 2, stride=2), nn.LeakyReLU(0.2),
            nn.ConvTranspose2d(256, 128, 2, stride=2), nn.LeakyReLU(0.2),
            nn.ConvTranspose2d(128, 64, 2, stride=2), nn.LeakyReLU(0.2),
            nn.ConvTranspose2d(64, 3, 2, stride=2), nn.Tanh(),
        )
    def forward(self, x): return self.up(x)

class PatchAutoencoder(nn.Module):
    def __init__(self):
        super().__init__()
        self.encoder = Encoder()
        self.decoder = Decoder()
    def forward(self, x):
        return self.decoder(self.encoder(x))
```

Listing 4: AE per-sample reconstruction error (`model_autoencoder.py`). `mean`: pixel MSE; `max_local`: max over 8×8 cell means.

```
def reconstruction_error_per_sample(model, x, aggregation="mean"):
    x_hat = model(x)
    err = (x - x_hat).pow(2).mean(dim=1, keepdim=True) # (B, 1, H, W)
    if aggregation == "mean":
        return err.view(x.size(0), -1).mean(dim=1)
    if aggregation == "max_local":
        B, _, H, W = err.shape
        cell = 8
        nc_h, nc_w = H // cell, W // cell
        err = err.view(B, 1, nc_h, cell, nc_w, cell).mean(dim=(3, 5))
        return err.view(B, -1).max(dim=1)[0]
```

Listing 5: ROC-based patch threshold: closest to (FPR=0, TPR=1) (`roc_threshold_optimal`, `model_autoencoder.py`).

```
def roc_threshold_optimal(errors, labels, positive_class=1):
    order = np.argsort(-errors)
    sorted_errors, sorted_labels = errors[order], (labels[order] == positive_class).
        astype(np.float64)
    n_pos, n_neg = sorted_labels.sum(), len(sorted_labels) - n_pos
    tpr = np.cumsum(sorted_labels) / n_pos
    fpr = np.cumsum(1.0 - sorted_labels) / n_neg
    thresh = np.concatenate([[sorted_errors[0]+1], sorted_errors, [sorted_errors
        [-1]-1]])
    thresh = (thresh[:-1] + thresh[1:]) / 2
    dist = fpr**2 + (1 - tpr)**2
    idx = np.argmin(dist)
    return float(thresh[idx]), float(tpr[idx]), float(fpr[idx])
```

Listing 6: Patient-level aggregation.

```
for patient in patients:
    patches = load_all_cropped_patches(patient)
    patch_scores = [model_score(p) for p in patches] # CNN prob or AE error
    contaminated = [1 if s > tau_patch else 0 for s in patch_scores]
    ratio = sum(contaminated) / len(contaminated)
    patient_pred = 1 if ratio > tau_patient else 0
```

Listing 7: Slurm training and comparison commands.

```
sbatch train_classifier.sbatch.sh
sbatch train_autoencoder.sbatch.sh
python compare_patch_classifiers.py
sbatch patient_cnn_fold.sbatch.sh
sbatch patient_autoencoder_fold.sbatch.sh
sbatch patient_plots.sbatch.sh
```